



Efficacy and safety of SGLT2 inhibitors with and without glucagon-like peptide 1 receptor agonists: a SMART-C collaborative meta-analysis of randomised controlled trials

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Summary

Background SGLT2 inhibitors and GLP-1 receptor agonists both improve cardiovascular and kidney outcomes in patients with type 2 diabetes. We sought to evaluate whether the benefits of SGLT2 inhibitors are consistent in patients receiving and not receiving GLP-1 receptor agonists.

Methods We conducted a collaborative meta-analysis of trials included in the SGLT2 Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium, restricted to participants with diabetes. Treatment effects from individual trials were obtained from Cox regression models and pooled using inverse variance weighted meta-analysis. The two main cardiovascular outcomes assessed included major adverse cardiovascular events (nonfatal myocardial infarction, nonfatal stroke, or cardiovascular death), and hospitalisation for heart failure or cardiovascular death. The main kidney outcomes assessed were chronic kidney disease progression ($\geq 40\%$ decline in estimated glomerular filtration rate [eGFR], kidney failure [eGFR < 15 mL/min/1.73 m², chronic dialysis, or kidney transplantation], or death due to kidney failure), and the rate of change in eGFR over time. Safety outcomes were also assessed.

Findings Across 12 randomised, double-blind, placebo-controlled trials, 3065 (4.2%) of 73 238 participants with diabetes were using GLP-1 receptor agonists at baseline. SGLT2 inhibitors reduced the risk of major adverse cardiovascular events in participants both receiving and not receiving GLP-1 receptor agonists (hazard ratio [HR] 0.81, 95% CI 0.63–1.03 vs 0.90, 0.86–0.94; p-heterogeneity=0.31). Effects on hospitalisation for heart failure or cardiovascular death (0.76, 0.57–1.01 vs 0.78, 0.74–0.82; p-heterogeneity=0.90) and chronic kidney disease progression (0.65, 0.46–0.94 vs 0.67, 0.62–0.72; p-heterogeneity=0.81) were also consistent regardless of GLP-1 receptor agonist use, as was the effect on the chronic rate of change in eGFR over time (heterogeneity=0.92). Fewer serious adverse events occurred with SGLT2 inhibitors compared with placebo, irrespective of GLP-1 receptor agonist use (relative risk 0.87, 95% CI 0.79–0.96 vs 0.91, 0.89–0.93; p-heterogeneity=0.41).

Interpretation The effects of SGLT2 inhibitors on cardiovascular and kidney outcomes are consistent regardless of the background use of GLP-1 receptor agonists. These findings suggest independent effects of these evidence-based therapies and support clinical practice guidelines recommending the use of these agents in combination to improve cardiovascular and kidney metabolic outcomes.

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Introduction

SGLT2 inhibitors reduce the risk of kidney failure, cardiovascular events, and death in patients with type 2 diabetes and those with heart failure or chronic kidney disease, regardless of diabetes status.¹ Consequently, consensus statements from professional diabetes, kidney, and cardiovascular organisations recommend that SGLT2 inhibitors be routinely offered to patients with type 2 diabetes with or at a high risk of atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease.^{2–4} Results from multiple large-scale randomised trials have

also shown that GLP-1 receptor agonists reduce the risk of cardiovascular events in people with type 2 diabetes.⁵ Therefore, prevailing type 2 diabetes treatment guidelines and society recommendations endorse the use of SGLT2 inhibitors or GLP-1 receptor agonists, or both, for individuals with type 2 diabetes with or at a high risk of atherosclerotic cardiovascular disease.⁶

The distinct mechanisms of action of these two classes of agents, their myriad effects on intermediate markers of cardiometabolic risk that differ between the classes, and the different types of events prevented have led to

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Research in context

Evidence before this study

Multiple large-scale randomised trials have shown that SGLT2 inhibitors and GLP-1 receptor agonists reduce the risk of cardiovascular events and improve chronic kidney disease outcomes in patients with type 2 diabetes. Given the different mechanisms of action of these two classes of agents as well as key differences in their effects on a myriad of intermediates of cardiovascular and kidney risk (eg, substantially greater effects on glycaemia and bodyweight with GLP-1 receptor agonists compared with SGLT2 inhibitors), there has been considerable interest in using SGLT2 inhibitors and GLP-1 receptor agonists in combination to reduce cardio-kidney-metabolic risk. We searched Medline and Embase from database inception to Jan 1, 2024, for randomised trials evaluating the effects of combination treatment with SGLT2 inhibitors and GLP-1 receptor agonists in people with type 2 diabetes, using the search terms "sodium glucose cotransporter 2 inhibitor", "glucagon-like peptide-1 receptor agonists" "type 2 diabetes mellitus", "clinical trial", "random allocation" and "double-blind method". We identified small, randomised trials of relatively short duration indicating that the combined use of SGLT2 inhibitors and GLP-1 receptor agonists improves cardiometabolic risk factors to a greater extent than either alone. However, there are few data on clinical outcomes. Because the event-driven randomised trials of each class of agent were conducted before the benefit of either was confirmed, the background use of GLP-1 receptor agonists in SGLT2 inhibitors trials was too infrequent in any single trial to evaluate the effects of SGLT2 inhibitors in patients receiving and not receiving GLP-1 receptor agonists. To address this evidence gap, we conducted a collaborative meta-analysis of randomised, double-blind, placebo-controlled, event-driven outcome trials within the SGLT2 Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium. We aimed to assess the effects of SGLT2 inhibitors on cardiovascular, kidney, and safety outcomes in patients with diabetes receiving and not receiving GLP-1 receptor agonists at baseline.

considerable interest in using SGLT2 inhibitors and GLP-1 receptor agonists in combination to further reduce cardiorenal event risk. For example, although SGLT2 inhibitors have substantial benefits on the incidence or worsening of heart failure and chronic kidney disease progression, GLP-1 receptor agonists might yield greater reductions in myocardial infarction and stroke.⁷ Guidelines therefore recognise the potential benefits of the combined use of both classes of agents.⁸

Since clinical trials of SGLT2 inhibitors and GLP-1 receptor agonists were conducted during approximately the same time period before the benefits of either therapy had been established, few participants in SGLT2 inhibitors outcome trials were receiving a GLP-1 receptor agonist. As a result, individual trials did not have sufficient power to examine the effects of SGLT2 inhibitors on clinical outcomes in patients receiving

Added value of this study

With 73 238 participants with diabetes from 12 large-scale randomised trials, including 3065 receiving GLP-1 receptor agonists at baseline, to the best of our knowledge, this collaborative meta-analysis represents the largest and most comprehensive assessment of the effects of SGLT2 inhibitors on cardiovascular, kidney, and safety outcomes in people with diabetes receiving and not receiving GLP-1 receptor agonists.

Compared with placebo, SGLT2 inhibitors reduced the risk of major adverse cardiovascular events (myocardial infarction, stroke, or cardiovascular death) by 11% and hospitalisation for heart failure or cardiovascular death by 23%, with consistent effects regardless of GLP-1 receptor agonist use at baseline. SGLT2 inhibitors also reduced the risk of chronic kidney disease progression (defined as a $\geq 40\%$ decline in estimated glomerular filtration rate, kidney failure, or death due to kidney failure) by 33% and attenuated the annual rate of decline in kidney function, irrespective of GLP-1 receptor agonist use at baseline. The effect of SGLT2 inhibitors on key safety outcomes, including serious adverse events, hypoglycaemia, and volume depletion, did not differ by baseline use of GLP-1 receptor agonists (all p -heterogeneity > 0.10).

Implications of all the available evidence

Pooled subgroup data from the totality of the worldwide large-scale placebo-controlled SGLT2 inhibitors trials suggest that the effects of SGLT2 inhibitors on cardiovascular, kidney, and safety outcomes are likely to be similar in patients prescribed a GLP-1 receptor agonist and those not, irrespective of indication for the use of SGLT2 inhibitors. These data suggest that the benefits of SGLT2 inhibitors are independent of GLP-1 receptor agonists, and provide support for the combined use of both classes of agent to optimise cardiovascular and kidney outcomes for patients with type 2 diabetes.

or not receiving GLP-1 receptor agonists. Accordingly, we conducted a meta-analysis of 12 completed trials enrolling patients with diabetes with or at a high risk of atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease to examine and compare the efficacy and safety of SGLT2 inhibitors with and without the use of GLP-1 receptor agonists at baseline in patients with diabetes.

Methods

Overview

The SGLT2 Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium (SMART-C) comprises completed, randomised, double-blind, placebo-controlled, event-driven trials that have assessed the effects of an SGLT2 inhibitor (including SGLT1 and SGLT2 inhibitors) on a primary clinical outcome in adults (aged ≥ 18 years),

restricted to trials with more than 500 participants in each group and with a follow-up of at least 6 months. SMART-C is led by an academic steering committee featuring key representatives from each of the collaborating trials. Analyses were conducted using standardised analytical techniques with harmonised endpoint definitions within each dataset using individual participant data, with results pooled across trials using two-stage meta-analysis.

Within the SMART-C framework, we conducted a collaborative meta-analysis using data from trials participating in SMART-C that enrolled patients with diabetes with or at high risk for atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease. These trials were identified through previously

published SMART-C systematic reviews and meta-analyses.¹⁹

Included trials

The present meta-analysis included 12 trials participating in SMART-C, with the analysis cohort restricted to trials exclusively enrolling participants with diabetes, and participants with diabetes enrolled in the following heart failure and chronic kidney disease trials: DAPA-CKD,¹⁰ EMPA-KIDNEY,¹¹ DAPA-HF,¹² EMPEROR-Reduced,¹³ EMPEROR-Preserved,¹⁴ and DELIVER.¹⁵ All trials excluded participants with type 1 diabetes except for EMPA-KIDNEY, which included 68 of these individuals. We excluded SOLOIST-WHF from this meta-analysis because this trial recruited participants

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	Used GLP-1 receptor agonist		Did not use GLP-1 receptor agonist	
	SGLT2 inhibitor	Placebo	SGLT2 inhibitor	Placebo
Diabetes at a high risk for atherosclerotic cardiovascular disease trials				
Participants, n	937	693	23 626	17 312
Age, years	63.0 (7.1)	62.8 (6.8)	63.7 (7.8)	63.8 (7.6)
Male	668 (71.3%)	486 (70.1%)	15 704 (66.5%)	11 174 (64.5%)
Female	269 (28.7%)	207 (29.9%)	7922 (33.5%)	6138 (35.5%)
History of cardiovascular disease	602 (64.2%)	410 (59.2%)	16 784 (71.0%)	11 044 (63.8%)
History of heart failure	85 (9.1%)	58 (8.4%)	3318 (14.0%)	2388 (13.8%)
Systolic blood pressure, mmHg	132.8 (15.5)	133.8 (15.2)	135.0 (15.5)	135.1 (15.6)
BMI, kg/m ²	34.9 (5.8)	35.0 (6.8)	31.5 (5.7)	31.5 (5.8)
HbA _{1c} , %	8.2 (1.0)	8.1 (1.0)	8.2 (1.0)	8.2 (1.1)
HbA _{1c} , mmol/mol	66.1	65.0	66.1	66.1
eGFR, mL/min/1.73 m ²	79.3 (20.5)	77.5 (19.8)	79.3 (19.8)	78.9 (19.4)
uACR ≥30 mg/g	312/925 (33.7%)	228/684 (33.3%)	8034/23 486 (34.2%)	5680/17 156 (33.1%)
RAS blockade use	807 (86.1%)	583 (84.1%)	19 063 (80.7%)	13 977 (80.7%)
Insulin use	477 (50.9%)	349 (50.4%)	10 788 (45.7%)	7781 (44.9%)
Chronic kidney disease trials				
Participants, n	635	640	9839	9817
Age, years	66.0 (8.8)	65.5 (9.1)	66.0 (9.4)	66.3 (9.3)
Male	418 (65.8%)	406 (63.4%)	5954 (60.5%)	5944 (60.5%)
Female	217 (34.2%)	234 (36.6%)	3885 (39.5%)	3873 (39.5%)
History of cardiovascular disease	296 (46.6%)	273 (42.7%)	4560 (46.3%)	4621 (47.1%)
History of heart failure	116 (18.3%)	112 (17.5%)	2248 (22.8%)	2252 (22.9%)
Systolic blood pressure, mmHg	138.0 (17.2)	137.4 (17.5)	139.1 (16.9)	139.4 (16.9)
BMI, kg/m ²	35.4 (6.8)	35.7 (7.6)	31.2 (6.4)	31.2 (6.3)
HbA _{1c} , %	8.1 (2.2)	8.2 (2.0)	8.2 (2.0)	8.2 (2.0)
HbA _{1c} , mmol/mol	65.0	66.1	66.1	66.1
eGFR, mL/min/1.73 m ²	44.4 (13.5)	43.0 (13.7)	43.8 (14.4)	43.6 (14.4)
uACR ≥30 mg/g	477 (75.1%)	463 (72.3%)	7804 (79.3%)	7799 (79.4%)
RAS blockade use	568 (89.4%)	583 (91.1%)	9057 (92.1%)	8960 (91.3%)
Insulin use	454 (71.5%)	496 (77.5%)	6024 (61.2%)	5893 (60.0%)
Heart failure trials				
Participants, n	88	72	4781	4798
Age, years	67.7 (8.9)	70.4 (9.1)	69.0 (9.7)	69.0 (9.7)
Male	65 (73.9%)	44 (61.1%)	3166 (66.2%)	3125 (65.1%)
Female	23 (26.1%)	28 (38.9%)	1615 (33.8%)	1673 (34.9%)

(Table continues on next page)

	Used GLP-1 receptor agonist		Did not use GLP-1 receptor agonist	
	SGLT2 inhibitor	Placebo	SGLT2 inhibitor	Placebo
(Continued from previous page)				
History of cardiovascular disease	62 (70.5%)	56 (77.8%)	3213 (67.2%)	3186 (66.4%)
History of heart failure	88 (100.0%)	72 (100.0%)	4781 (100.0%)	4798 (100.0%)
Systolic blood pressure, mmHg	127.9 (17.4)	126.9 (16.7)	127.4 (16.5)	127.5 (16.6)
BMI, kg/m ²	31.3 (6.8)	31.3 (7.1)	29.9 (6.0)	30.0 (6.0)
HbA _{1c} , %	8.0 (1.5)	7.5 (1.7)	7.4 (1.6)	7.4 (1.6)
HbA _{1c} , mmol/mol	63.9	58.5	57.4	57.4
eGFR, mL/min/1.73 m ²	53.3 (20.3)	56.2 (20.2)	60.8 (20.4)	61.0 (20.4)
uACR ≥30 mg/g*	26/43 (60.5%)	14/35 (40.0%)	1223/2350 (52.0%)	1201/2366 (50.8%)
RAS blockade use	67 (76.1%)	56 (77.8%)	4119 (86.2%)	4082 (85.1%)
Insulin use	53 (60.2%)	44 (61.1%)	1286 (26.9%)	1333 (27.8%)
Data are mean (SD) or n (%). eGFR=estimated glomerular filtration rate. RAS=renin-angiotensin system. uACR=urine albumin to creatinine ratio. *Only available in EMPEROR-Reduced and EMPEROR-Preserved trials.				
Table: Characteristics of participants stratified by study population and use of GLP-1 receptor agonists at baseline				

with recent acute decompensated heart failure who were followed up for a short duration that precluded the assessment of effects on chronic kidney disease progression.¹⁶ All trials were approved by relevant ethics committees, and all participants provided written informed consent for research participation.

Outcomes

The two main cardiovascular outcomes assessed were first, major adverse cardiovascular events, defined as a composite of nonfatal myocardial infarction, nonfatal stroke, or cardiovascular death; and second, hospitalisation for heart failure or cardiovascular death. The two kidney outcomes assessed included first, a composite kidney outcome using a standardised definition of percentage decline in estimated glomerular filtration rate (eGFR) and kidney failure across the trials: a 40% or more decline in eGFR, kidney failure (eGFR <15 mL/min/1.73 m², chronic dialysis, or kidney transplantation), or death due to kidney failure; and second, the annual rate of change in eGFR (eGFR slope), both total (calculated from the visit at which they were randomly assigned) and chronic (from the first on-treatment eGFR measurement after random assignment).

All cardiovascular outcomes were adjudicated by expert masked adjudication committees. We evaluated the safety outcomes that could be affected by combination treatment with SGLT2 inhibitors and GLP-1 receptor agonists: any serious adverse event, adverse events due to hypoglycaemia, or adverse events due to volume depletion. Because adverse event reporting differed across the trials (ie, some trials collected serious and non-serious adverse events, whereas others only collected serious events), we primarily focused on serious adverse events, since this outcome was available in all trials using common serious adverse event criteria for reporting, but

we also assessed investigator-reported hypoglycaemia and volume depletion as reported in each trial individually.

Data analysis

We undertook a two-stage meta-analysis using harmonised analytical approaches and endpoint definitions across the individual trials as aforementioned. Investigators from each trial analysed the treatment effects and safety events within each trial using individual participant data. In trials that included patients with and without diabetes, analyses were restricted to patients with diabetes, since GLP-1 receptor agonists were only indicated for use in these individuals at the time the trials were conducted. We compared the characteristics (age, sex, history of cardiovascular disease, heart failure, systolic blood pressure, BMI, HbA_{1c}, eGFR, urine albumin to creatinine ratio, renin-angiotensin system blockade use, and insulin use) of participants receiving and not receiving GLP-1 receptor agonists at baseline across the three studied populations (patients with diabetes with or at a high risk of atherosclerotic cardiovascular disease, heart failure, and chronic kidney disease). We calculated the pooled weighted mean and SDs for continuous variables using the weighted inverse variance method and pooled categorical variables across trials by treatment allocation and baseline GLP-1 receptor agonists use.

Details on the treatment effects on efficacy outcomes by baseline use of GLP-1 receptor agonists in each trial were obtained from Cox regression models expressed as hazard ratios (HRs) with corresponding 95% CIs, with covariate stratification as prespecified in individual trials and without adjustment for differences in characteristics by baseline GLP-1 receptor agonist use. For safety outcomes, we obtained HRs and 95% CIs where time-to-event data were available, or used risk ratios based on the number of participants with an event where time-to-event

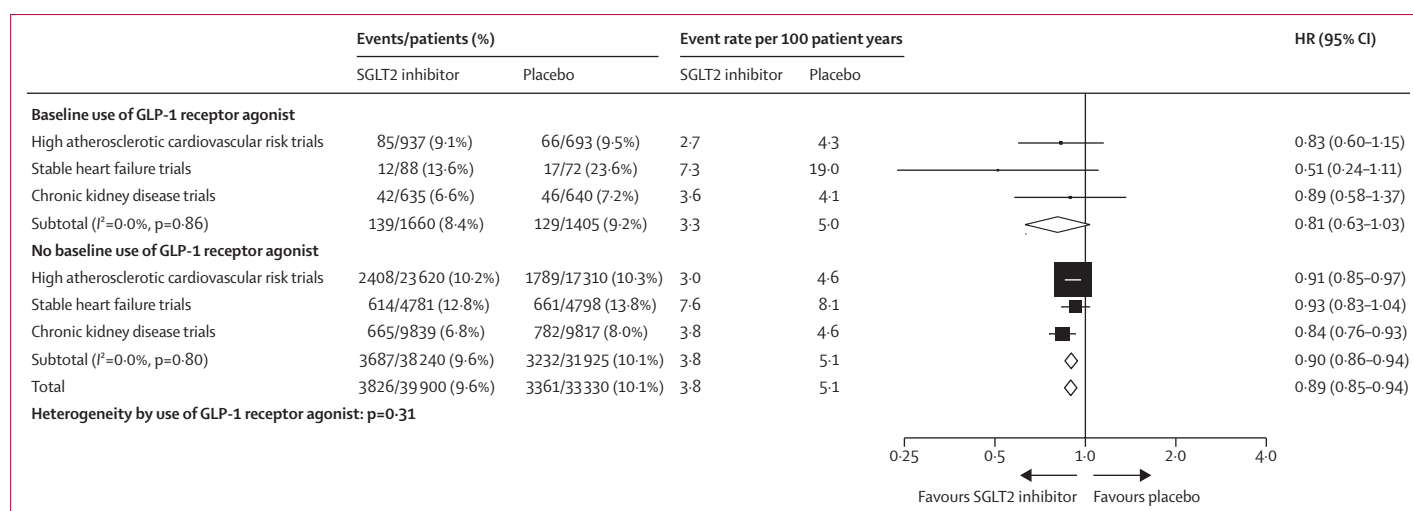


Figure 1 Effect of SGLT2 inhibitors on major adverse cardiovascular events (myocardial infarction, stroke, or cardiovascular death) with and without the use of GLP-1 receptor agonists at baseline
HR=hazard ratio.

data were not available. Summary effect estimates, both overall and stratified by trial population and baseline GLP-1 receptor agonist use, were obtained by pooling log-transformed HRs and risk ratios using inverse-variance weighted averages. Heterogeneity of treatment effects by baseline use of GLP-1 receptor agonists was then assessed using p values for heterogeneity obtained from random-effects meta-regression with restricted maximum likelihood.

A two-slope mixed-effects linear spline model with unstructured covariance matrix was used for analyses of the eGFR slope.^{10,11,17} The chronic and total eGFR slopes were defined on the basis of this two-slope model, with a knot generally placed at the first eGFR value after random assignment in each trial, typically within the first 8 weeks after random assignment. eGFR slope analyses were restricted to on-treatment eGFR values because of the increase in eGFR after SGLT2 inhibitor discontinuation that could theoretically affect eGFR slope estimates. Models included two-way and three-way interaction terms between randomised treatment, GLP-1 receptor agonist use, and two-slope linear spline in follow-up time to account for differences between GLP-1 receptor agonist subgroups in the effect of the treatment on the mean eGFR trajectories. The modelled estimates of total and chronic GFR slopes, expressed as the annual rate of change in eGFR (mL/min/1.73 m² per year) were then meta-analysed across all trials using the same approach as described earlier. All trial-level meta-analyses were performed with R version 4.3.1.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

We included participants from 12 randomised trials, of whom 73 238 (82.1%) of 89 183 had diabetes. Details of these 12 trials are summarised in the appendix (p 2). Among the 73 238 participants with diabetes, 3065 (4.2%) were using GLP-1 receptor agonists at baseline. GLP-1 receptor agonist use varied across all trials, with a range of 2.8–4.4% in the atherosclerotic cardiovascular disease trials, 4.2–11.1% in the chronic kidney disease progression trials, and 1.0–2.2% in the heart failure trials (appendix p 2). Across the studied populations, the characteristics of participants receiving and not receiving GLP-1 receptor agonists were similar in many aspects (table). However, those receiving GLP-1 receptor agonists were more likely to be male, have a higher BMI, and be receiving insulin (table).

Across all 12 trials, 7187 (9.8%) participants had a major adverse cardiovascular event, including 268 (3.7%) who were receiving GLP-1 receptor agonist and 6919 (9.3%) who were not receiving a GLP-1 receptor agonist at baseline. The incidence of major adverse cardiovascular events was highest in the heart failure trials compared with all other trials, driven by cardiovascular death (figure 1). The effect of SGLT2 inhibitors on major adverse cardiovascular events (HR 0.89, 95% CI 0.85–0.94) was consistent in participants receiving (0.81, 0.63–1.03) and not receiving (0.90, 0.86–0.94) a GLP-1 receptor agonist (p -heterogeneity=0.31; figure 1).

Overall, 6260 (8.5%) participants were hospitalised for heart failure or died due to cardiovascular disease, of whom 211 (3.4%) were receiving and 6049 (9.6%) were not receiving a GLP-1 receptor agonist at baseline. The incidence of hospitalisation for heart failure or cardiovascular death was highest in the heart failure trials (figure 2). The effect of SGLT2 inhibitors on hospitalisation for heart failure or cardiovascular death (HR 0.77, 95% CI

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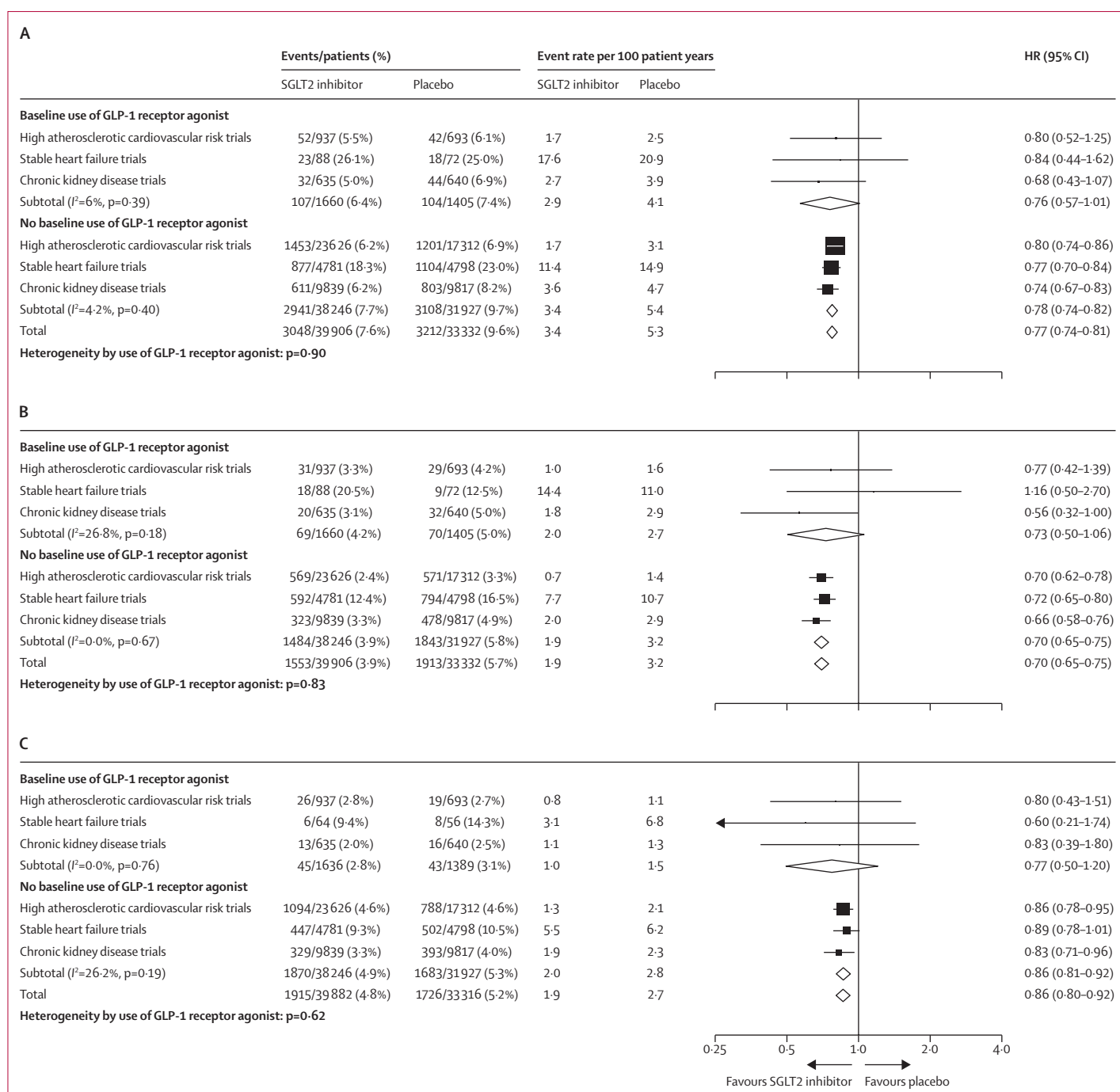


Figure 2: Effect of SGLT2 inhibitors on (A) hospitalisation for heart failure or cardiovascular death, (B) hospitalisation for heart failure, and (C) cardiovascular death, with and without the use of GLP-1 receptor agonists at baseline
HR=hazard ratio.

0.74–0.81) was consistent in participants receiving GLP-1 receptor agonists (0.76, 0.57–1.01) and those not receiving GLP-1 receptor agonists (0.78, 0.74–0.82; p -heterogeneity 0.90; figure 2A). The effect of SGLT2 inhibitors on hospitalisation for heart failure was similarly consistent regardless of GLP-1 receptor agonist use at baseline (0.73, 0.50–1.06 vs 0.70, 0.65–0.75;

p -heterogeneity=0.83; figure 2B) as was the effect on cardiovascular death (0.77, 0.50–1.20 vs 0.86, 0.81–0.92; p -heterogeneity=0.62; figure 2C).

The incidence of the composite kidney outcome was highest in the chronic kidney disease trials, with 3294 (4.5%) participants having chronic kidney disease progression, including 130 (3.9%) who were receiving

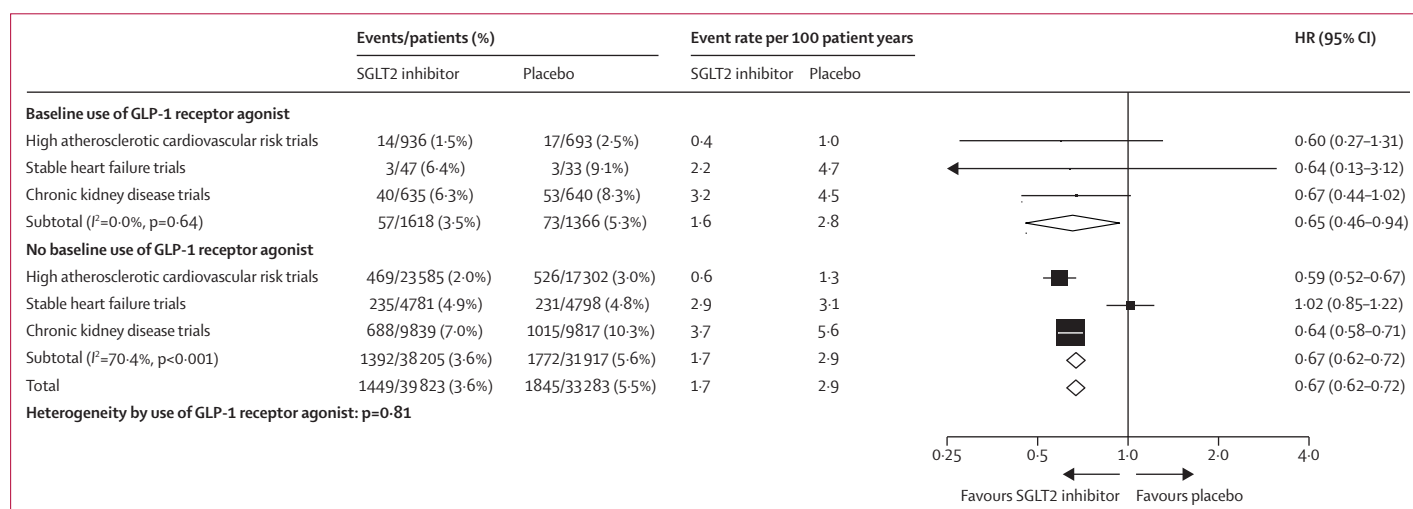


Figure 3: Effect of SGLT2 inhibitors on chronic kidney disease progression ($\geq 40\%$ decline in estimated glomerular filtration rate, kidney failure, or death due to kidney failure), with and without the use of GLP-1 receptor agonists at baseline
HR=hazard ratio.

and 3164 (96.1%) who were not receiving a GLP-1 receptor agonist at baseline (figure 3). SGLT2 inhibitors reduced the risk of chronic kidney disease progression (HR 0.67, 95% CI 0.62–0.72) in participants receiving and not receiving GLP-1 receptor agonists (0.65, 0.46–0.94 vs 0.67, 0.62–0.72; p -heterogeneity=0.81; figure 3).

The consistent benefits of SGLT2 inhibitors on clinical outcomes in participants being prescribed or not being prescribed a GLP-1 receptor agonist at baseline were observed across the studied populations. Participants with type 1 diabetes comprised less than 0.1% of the no GLP-1 receptor agonist subgroup, with none receiving GLP-1 receptor agonists at baseline. As a result, treatment effects estimates were identical when excluding these individuals. The treatment effects on cardiovascular and kidney outcomes by individual trial are summarised in the appendix (pp 3–6).

SGLT2 inhibitors attenuated the annual rate of decline in eGFR as measured by chronic and total eGFR slope, regardless of baseline GLP-1 receptor agonist use (figure 4). The chronic eGFR slope was improved by 1.32 mL/min/1.73 m² per year (95% CI 0.98–1.66) in participants receiving GLP-1 receptor agonists and by 1.21 mL/min/1.73 m² per year (1.14–1.28) in participants not receiving GLP-1 receptor agonists at baseline (p -heterogeneity=0.55; figure 4A). In relative terms, this corresponded to a 57% (54–60) reduction in the chronic slope, which was observed similarly in participants receiving and not receiving GLP-1 receptor agonists at baseline (60%, 45–76 vs 57%, 53–60; p -heterogeneity=0.43; figure 4A). Similar results were observed for the total eGFR slope in participants receiving and not receiving GLP-1 receptor agonists at baseline (absolute difference 0.67 mL/min/1.73 m² per year, 0.41–0.94 vs 0.66 mL/min/1.73 m² per year,

0.61–0.71; p -heterogeneity=0.92; figure 4B). This translated into a 30% (28–32) relative reduction in the total slope overall, which was again observed regardless of baseline GLP-1 receptor agonists use (32%, 19–44 vs 30%, 28–32; p -heterogeneity=0.82; figure 4B). The treatment effects on eGFR slope in individual trials are summarised in the appendix (pp 7–8).

A total of 5822 (7.9%) participants died during follow up, of whom 160 (2.7%) were receiving and 5662 (97.3%) were not receiving GLP-1 receptor agonists at baseline. The effect of SGLT2 inhibitors on all-cause mortality (HR 0.89, 95% CI 0.84–0.93) was consistent regardless of GLP-1 receptor agonists use at baseline (0.82, 0.60–1.13 vs 0.89, 0.84–0.94; p -heterogeneity=0.63; figure 5A).

Fewer serious adverse events were observed with SGLT2 inhibitors compared with placebo, regardless of GLP-1 receptor agonist use at baseline (relative risk 0.87, 95% CI 0.79–0.96 vs 0.91, 0.89–0.93; p -heterogeneity=0.41; figure 5B). Overall, SGLT2 inhibitors did not increase the risk of hypoglycaemia (relative risk 0.96; 95% CI 0.88–1.04), an absence of effect that was consistent by baseline use of GLP-1 receptor agonists (p -heterogeneity=0.13; figure 5C). SGLT2 inhibitors increased the risk of volume depletion (relative risk 1.21, 95% CI 1.10–1.32), an effect which was also consistent irrespective of GLP-1 receptor agonist use (p -heterogeneity=0.55). The treatment effects on safety outcomes by individual trial are summarised in the appendix (p 9).

Discussion

This collaborative meta-analysis of 12 large placebo-controlled outcome trials of SGLT2 inhibitors involving more than 70000 participants with diabetes, of whom more than 3000 were receiving a GLP-1 receptor agonist at baseline, represents the largest and most comprehensive

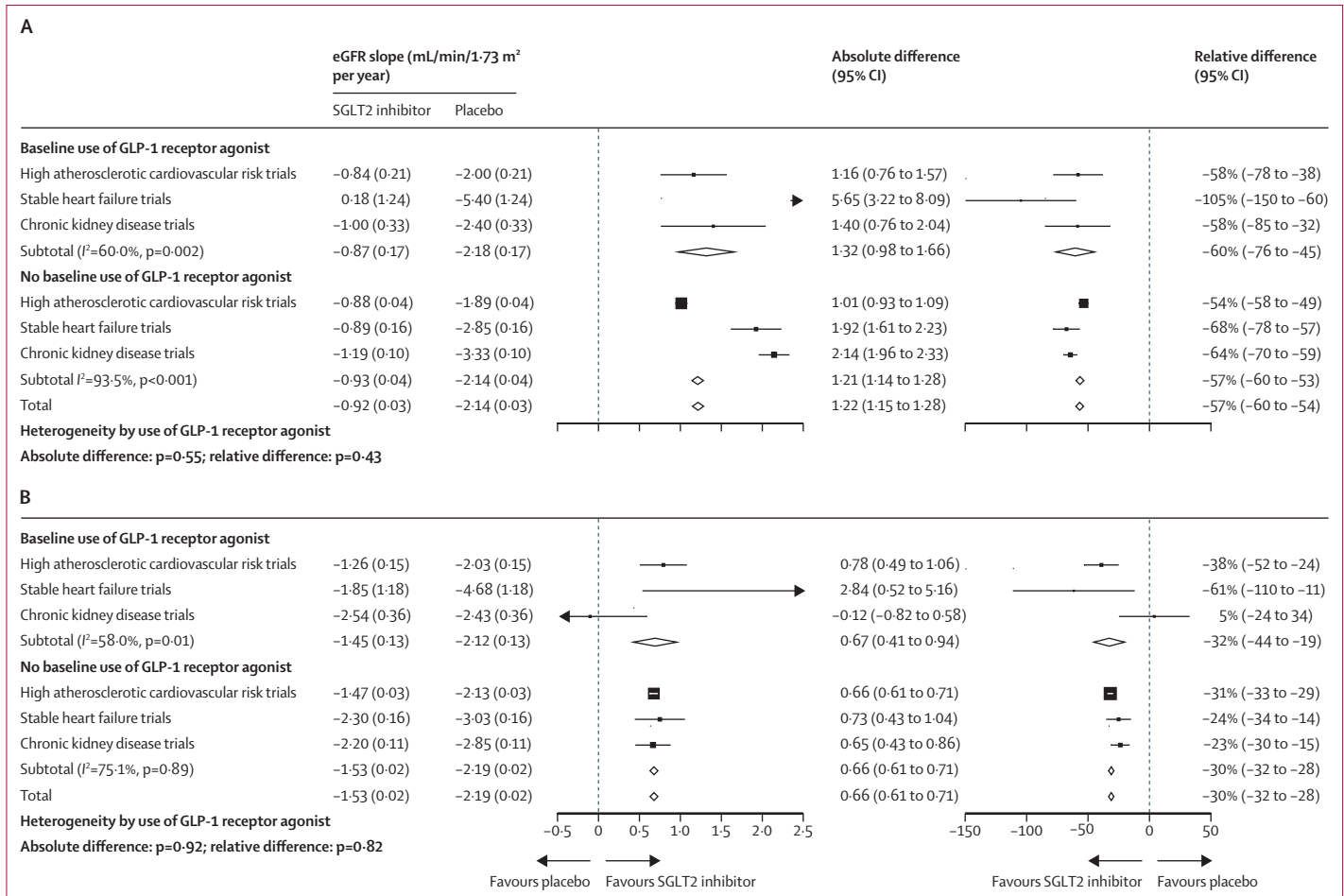


Figure 4: Effects of SGLT2 inhibitors on (A) chronic eGFR slope and (B) total eGFR slope, with and without the use of GLP-1 receptor agonists at baseline
eGFR=estimated glomerular filtration rate.

evaluation of the effects of SGLT2 inhibitors on clinical outcomes, with and without GLP-1 receptor agonist use. The smaller proportion of participants receiving GLP-1 receptor agonists at baseline is reflected in the wider 95% CIs for treatment effects in this subgroup. Notwithstanding the greater imprecision, the data (including the eGFR slope) clearly indicate that reductions in cardiovascular, kidney, and mortality outcomes with SGLT2 inhibitors are consistent, regardless of GLP-1 receptor agonist use at baseline. Additionally, the safety and tolerability of SGLT2 inhibitors appeared consistent in those receiving and not receiving a GLP-1 receptor agonist. These data provide support for the combined use of SGLT2 inhibitors and GLP-1 receptor agonists to optimise cardiovascular and kidney outcomes for patients with type 2 diabetes.

Contemporary guidelines and society consensus recommendations recognise the benefits of combination treatment with SGLT2 inhibitors and GLP-1 receptor agonists.^{2,3,6} At the same time, the rationale for their combined use has been largely based on their different mechanisms of action and greater benefits observed on

cardiometabolic risk factors observed in small trials of a short duration, although results have not always been consistent.¹⁸⁻²¹ Our findings are consistent with current clinical practice guideline from major diabetes, cardiovascular, and nephrology organisations recommending the preferential use of both SGLT2 inhibitors and GLP-1 receptor agonists in people with type 2 diabetes having an indication for each therapy.²²

The consistency of SGLT2 inhibitor treatment effects by baseline GLP-1 receptor agonist use across the studied populations is of considerable importance, given the rapidly expanding cardiovascular indications for GLP-1 receptor agonist therapy, including in people without diabetes. The SELECT and STEP-HF trials reported that the GLP-1 receptor agonist semaglutide reduces cardiovascular events in individuals with obesity and established cardiovascular disease, and improves functional status in patients with heart failure without diabetes.^{23,24} Thus, the combined use of SGLT2 inhibitors and GLP-1 receptor agonists across the spectrum of cardio-kidney-metabolic diseases is likely to increase substantially. Clinicians and patients might therefore be reassured by

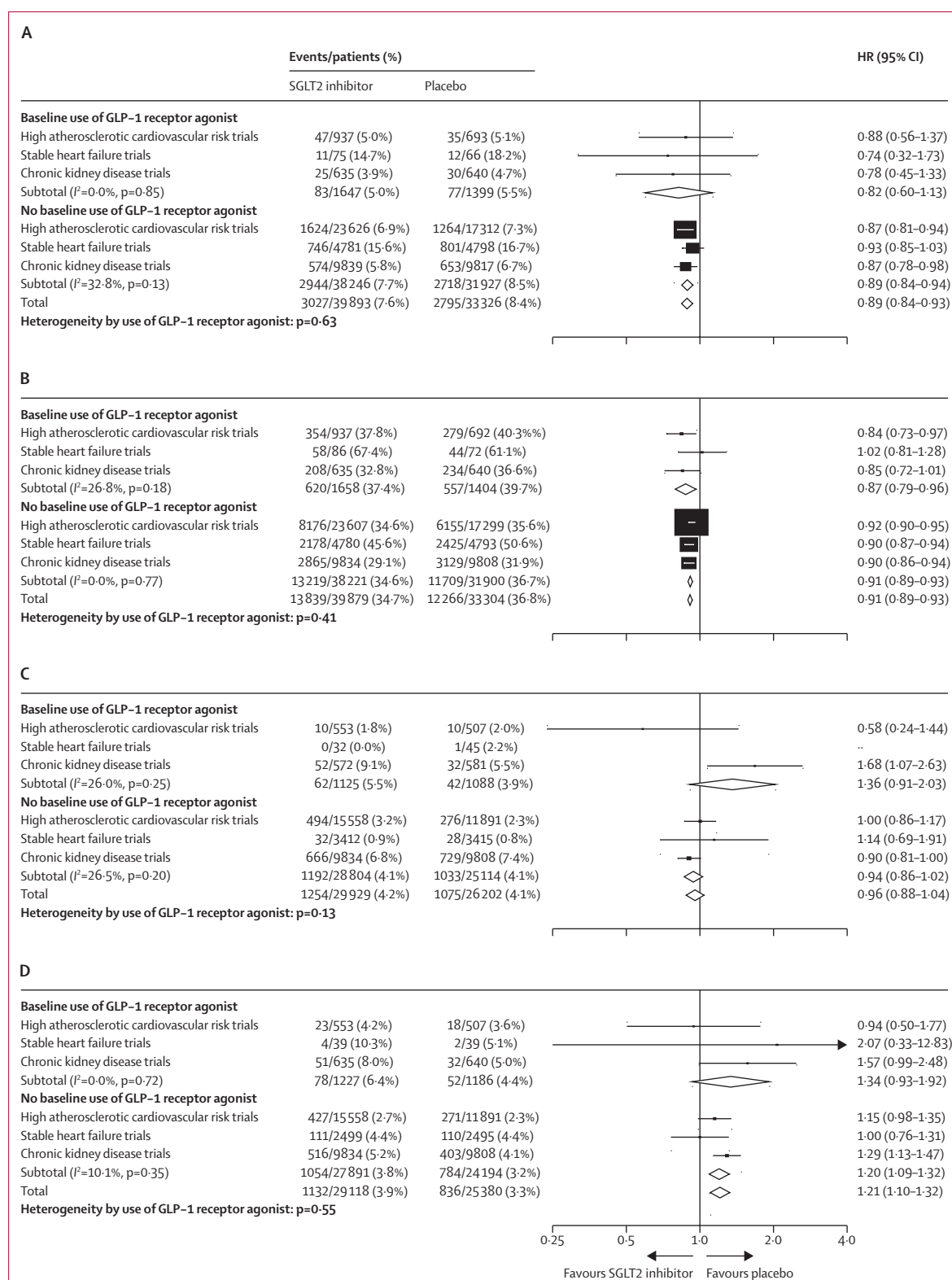


Figure 5: Effects of SGLT2 inhibitors on (A) all-cause mortality, (B) serious adverse events, (C) hypoglycaemia, and (D) volume depletion, with and without the use of GLP-1 receptor agonists at baseline

HR=hazard ratio.

the finding that SGLT2 inhibitors appear to be generally well tolerated in patients receiving and not receiving GLP-1 receptor agonists across a broad population of underlying cardio-kidney-metabolic diseases.

The cardiovascular and kidney protective effects of SGLT2 inhibitors and GLP-1 receptor agonists are likely to be explained through distinct mechanistic pathways. Both classes of agents reduce cardiometabolic risk factors such as HbA_{1c}, bodyweight, blood pressure, and albuminuria, although the effects of GLP-1 receptor agonists on glycaemia and bodyweight are substantially greater than those of SGLT2 inhibitors. However, these effects are unlikely to fully explain the magnitude of benefit observed on the clinical outcomes for either class.^{25,26} Although SGLT2 inhibitors reduce measures of glomerular pressure, which might contribute to their kidney protective effects,^{27,28} effects on tubular energy expenditure, nutrient deprivation signalling, autophagy, and inflammation are also likely to be important.²⁹ In contrast, GLP-1 receptor agonists act to stabilise atherosclerotic plaques as shown in various experimental and clinical studies, and suppress pro-inflammatory and pro-fibrotic pathways, with the latter pathway suppression potentially contributing to kidney benefits.³⁰ Thus, these two classes of agents have distinct domains, clinically and biologically, such that the beneficial cardiovascular and kidney effects are likely to be complementary.

The clear and separate benefit of SGLT2 inhibitors on the composite kidney endpoint in patients receiving and not receiving GLP-1 receptor agonists is of particular relevance in the context of the FLOW trial,³¹ the only dedicated chronic kidney disease progression trial of a GLP-1 receptor agonist in patients with type 2 diabetes. The FLOW trial reported a statistically significant 24% reduction in cardiovascular and kidney disease progression or cardiovascular or kidney failure related death with semaglutide compared with placebo.³¹ Our finding that SGLT2 inhibitors reduced the risk of the composite kidney endpoint, regardless of GLP-1 receptor agonist use, is strengthened by the analyses of the eGFR slope, a validated surrogate outcome for chronic kidney disease progression,³² which provides substantially increased power to examine treatment effects across subgroups of patients. The consistent benefits on the chronic and total slopes and the composite kidney endpoint in patients receiving and not receiving GLP-1 receptor agonists provides the strongest evidence yet for the efficacy of SGLT2 inhibitors, when added to GLP-1 receptor agonists, to reduce the risk of kidney failure in patients with type 2 diabetes.

Although the benefits of SGLT2 inhibitors and GLP-1 receptor agonists are well established, the uptake of these therapies is suboptimal across most health-care settings worldwide.^{33,34} Ensuring equitable access to SGLT2 inhibitors and GLP-1 receptor agonists will be important to ensure those who are likely to benefit most from combination cardio-kidney-metabolic therapies are able

to receive them, with cost being a major barrier to access across many jurisdictions. Economic evaluation of the combined use of SGLT2 inhibitors and GLP-1 receptor agonists across different health-care settings might also provide valuable information. Finally, understanding and addressing the barriers to widespread implementation will be crucial to realising the full potential of these transformative therapies.

There are some limitations to acknowledge when interpreting our findings. The number of outcomes among the patients using GLP-1 receptor agonists was much lower than those not using GLP-1 receptor agonists. As a result, the number of composite kidney endpoints in participants receiving GLP-1 receptor agonists was low. However, the use of an eGFR slope provided an opportunity to evaluate the consistency of effects on chronic kidney disease progression, with findings that were entirely consistent with the composite endpoint. The estimated treatment effects were not derived from one-stage meta-analysis. However, in lieu of pooled individual participant data, our use of a harmonised analytical approach and outcome definitions within a two-stage meta-analysis framework is consistent with the approach taken with other large cardio-metabolic consortia,^{32,35} and a notable strength of this work. We did not evaluate whether the effects of GLP-1 receptor agonists on clinical outcomes were similar with and without SGLT2 inhibitors, although subgroup data from two large GLP-1 receptor agonist trials suggest that the effects of GLP-1 receptor agonists are similar regardless of SGLT2 inhibitor use.³⁶ We only analysed treatment effects by baseline GLP-1 receptor agonist use without the ability to perform time-updated analyses on the basis of GLP-1 receptor agonist use at any timepoint, although such analyses introduce potential bias by the way of post-randomisation subgroups. Our analyses group all GLP-1 receptor agonists together; however, there might be important differences in efficacy or safety, or both, among different agents within the class. Finally, this work is not a definitive randomised test of whether effects are independent and additive, and it is unable to address the question of whether the combined initiation of GLP-1 receptor agonists and SGLT2 inhibitors is beneficial. Direct support for the additive benefits of combination therapy with both drug classes would require a dedicated clinical outcome trial.

In summary, the effects of SGLT2 inhibitors in reducing cardiovascular and kidney outcomes were similar in patients receiving and not receiving GLP-1 receptor agonists at baseline. These findings suggest independent effects of these evidence-based therapies and support clinical practice guidelines recommending the use of both these agents to optimise cardiovascular and kidney outcomes.

Contributors

BLN and HJLH conceived the meta-analysis and developed its design with EMA, RAF, and NJ. SEI and CW provided data from the EMPA-REG OUTCOME trial. BN, VP, MJJ, and KWM provided data from the

CANVAS Program and CREDENCE trials. SDW and MSS provided data from the DECLARE-TIMI 58 trial. DZIC, C-CL, and DKM provided data from the VERTIS trial. DCW and HJLH provided data from the DAPA-CKD trial. JJVM provided data from the DAPA-HF trial. DLB and MS provided data from the SCORED trial. SDA, JB, MP, and FZ provided data from the EMPEROR-Reduced and EMPEROR-Preserved trials. SDS and MV provided data from the DELIVER trial. WGH and NS provided data from the EMPA-KIDNEY trial. Data extraction and preparation was performed by EMA and RAF. Statistical analyses were done by RAF. EMA, BLN, and HJLH wrote the first draft of the manuscript. All authors contributed to the data interpretation and manuscript revisions. All authors had access to all the data. BLN, EMA, RAF, and HJLH accessed and verified the data. EMA, BLN, and HJLH take final responsibility for the decision to submit for publication.

Declaration of interests

BLN reports fees for travel support, advisory boards, scientific presentations, and steering committee roles from AstraZeneca, Alexion, Bayer, Boehringer and Ingelheim, Cambridge Healthcare Research, Cornerstone Medical Education, Janssen, the limbic, Medscape, Novo Nordisk, and Travers Therapeutics with all honoraria paid to The George Institute for Global Health. RAF has received studentship awards from the HDR-UK-Turing Wellcome Programme in Health Data Science. SDA reports grants and personal fees from Vifor and Abbott Vascular; and personal fees for consultancies, trial committee work or lectures, or a combination, from Actimed, Amgen, Astra Zeneca, Bayer, Boehringer Ingelheim, Brahms, Cardiac Dimensions, Cardior, Cordio, CVRx, Cytokinetics, Edwards, Farraday Pharmaceuticals, GSK, HeartKinetics, Impulse Dynamics, Occlutech, Pfizer, Regeneron, Repairon, Scirent, Sensible Medical, Servier, Vectorious, and V-Wave; and is the named co-inventor of two patent applications regarding MR-proANP (patent numbers DE 102007010834 and DE 102007022367), but he does not benefit personally from the related issued patents. DLB is on an advisory board for Angiowave, Bayer, Boehringer Ingelheim, CellProthera, Cereno Scientific, Elsevier Practice Update Cardiology, High Enroll, Janssen, Level Ex, McKinsey, Medscape Cardiology, Merck, MyoKardia, NirvaMed, Novo Nordisk, PhaseBio, PLx Pharma, Stasys; is on the board of directors for American Heart Association New York City, Angiowave (stock options), Bristol Myers Squibb (stock), DRS. LINQ (stock options), High Enroll (stock); is a consultant for Broadview Ventures, GlaxoSmithKline, Hims, SFJ, Youngene; is on a data monitoring committee for Acesion Pharma, Assistance Publique-Hôpitaux de Paris, Baim Institute for Clinical Research (formerly Harvard Clinical Research Institute, for the PORTICO trial, funded by St Jude Medical, now Abbott), Boston Scientific (chair for the PEITHO trial), Cleveland Clinic, Contego Medical (chair for the PERFORMANCE 2 trial), Duke Clinical Research Institute, Mayo Clinic, Mount Sinai School of Medicine (for the ENVISAGE trial, funded by Daiichi Sankyo; for the ABILITY-DM trial, funded by Concept Medical; and for ALLAY-HF, funded by Alleviant Medical), Novartis, Population Health Research Institute, Rutgers University (for the National Institutes for Health-funded MINT trial); receives honoraria from the American College of Cardiology (senior associate editor for Clinical Trials and News for ACC.org; chair for the American College of Cardiology Accreditation Oversight Committee), Arnold and Porter law firm (work related to Sanofi and Bristol-Myers Squibb clopidogrel litigation), Baim Institute for Clinical Research (formerly Harvard Clinical Research Institute; for the RE-DUAL PCI clinical trial steering committee funded by Boehringer Ingelheim; and the AEGIS-II executive committee funded by CSL Behring), Belvoir Publications (editor in chief for Harvard Heart Letter), Canadian Medical and Surgical Knowledge Translation Research Group (clinical trial steering committees), CSL Behring (American Heart Association lecture), Cowen and Company, Duke Clinical Research Institute (clinical trial steering committees, including for the PRONOUNCE trial, funded by Ferring Pharmaceuticals), HMP Global (editor in chief for the Journal of Invasive Cardiology), Journal of the American College of Cardiology (guest editor and associate editor), K2P (co-chair for interdisciplinary curriculum), Level Ex, Medtelligence/ReachMD (Continuing Medical Education steering committees), MJH Life Sciences, Oakstone CME (course director for Comprehensive Review of Interventional

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Data sharing

Detailed information on data sharing policies for the individual trials included within SMART-C are available in the primary publication of each relevant trial.

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